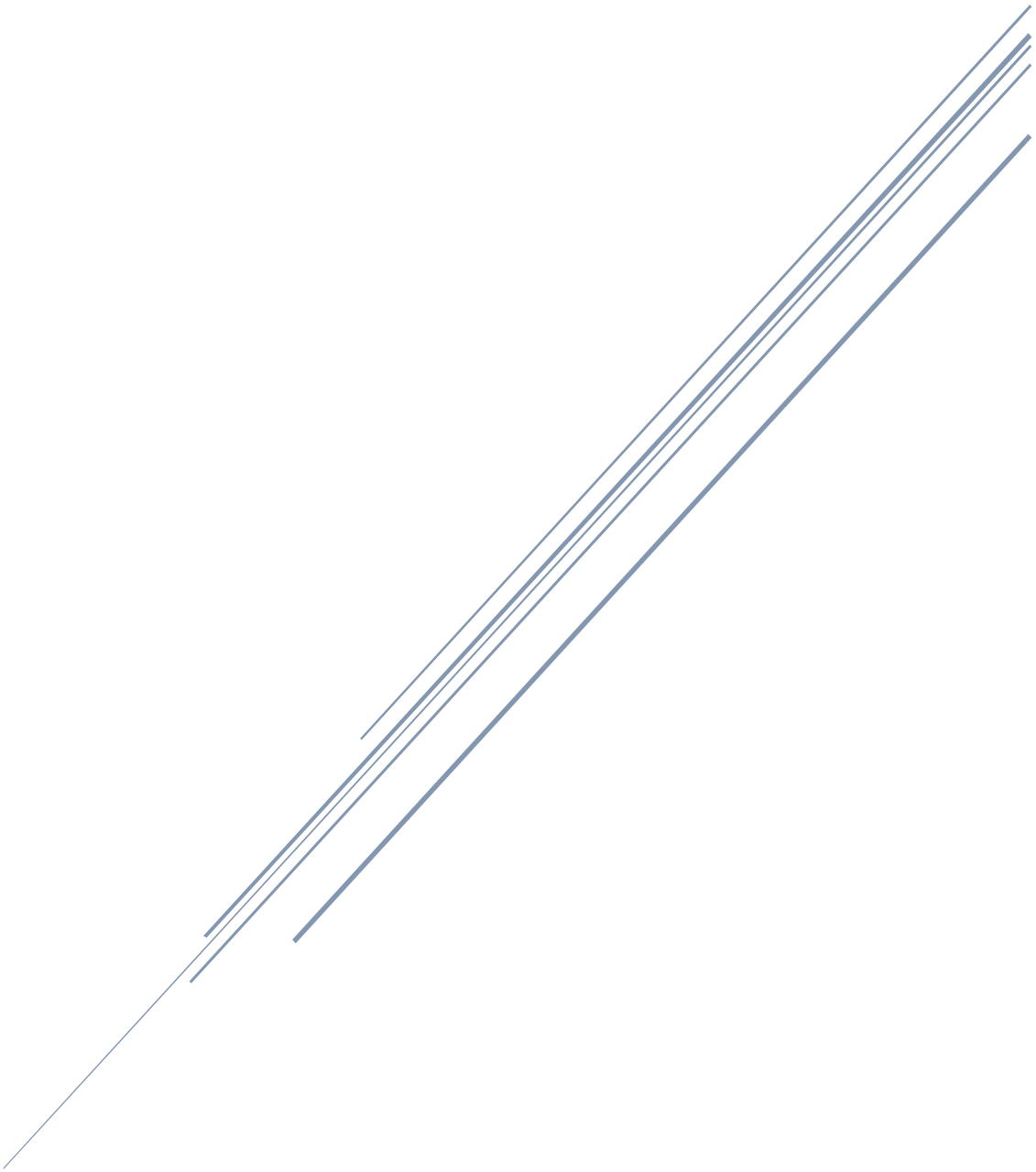
INDIAN KNOWLEDGE SYSTEM PROJECT

AYURVEDIC CLASSIFICATION AS A RULE-BASED EXPERT SYSTEM



THEEM COLLEGE OF ARTS COMMERCE AND SCIENCE
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Ayurvedic Classification – Rule-Based Expert System

Title: Ayurvedic Classification as a Rule-Based Expert System

Introduction

Ayurvedic Classification as a Rule-Based Expert System is a **B.Sc. Computer Science project** that demonstrates how traditional Indian knowledge can be represented and processed using concepts from **Artificial Intelligence and Expert Systems**.

Ayurveda is an ancient Indian knowledge system that uses the three doshas—**Vata, Pitta, and Kapha**—to describe different body characteristics and tendencies. In this project, selected Ayurvedic characteristics such as **body type, appetite, activity level, sleep pattern, and response to weather** are converted into computer-readable facts and rules.

The system uses a **Rule-Based Expert System**, where predefined **IF–THEN rules** represent the knowledge. For example, *IF appetite is strong THEN increase the Pitta score*. The system applies these rules to the user's inputs, calculates the scores of Vata, Pitta, and Kapha, and identifies the dominant classification.

From a **B.Sc. Computer Science perspective**, this project demonstrates important concepts such as **knowledge representation, rule-based reasoning, inference engines, classification, decision-making, and Python programming**. It also shows how traditional Indian Knowledge Systems (IKS) can be connected with modern Computer Science techniques.

The project is intended as an **educational demonstration of rule-based classification** and is not a medical diagnosis or treatment system.

Problem Statement

Ayurveda is a traditional Indian knowledge system that classifies individuals and health-related conditions using concepts such as **Vata, Pitta, and Kapha** (the three doshas). In this project, the Ayurvedic classification concept is mapped to a **Rule-Based Expert System** in Computer Science.

The system accepts simple user characteristics such as **body tendency, appetite, activity level, sleep pattern, and response to weather**. It applies predefined **IF–THEN rules** and calculates which dosha has the highest score.

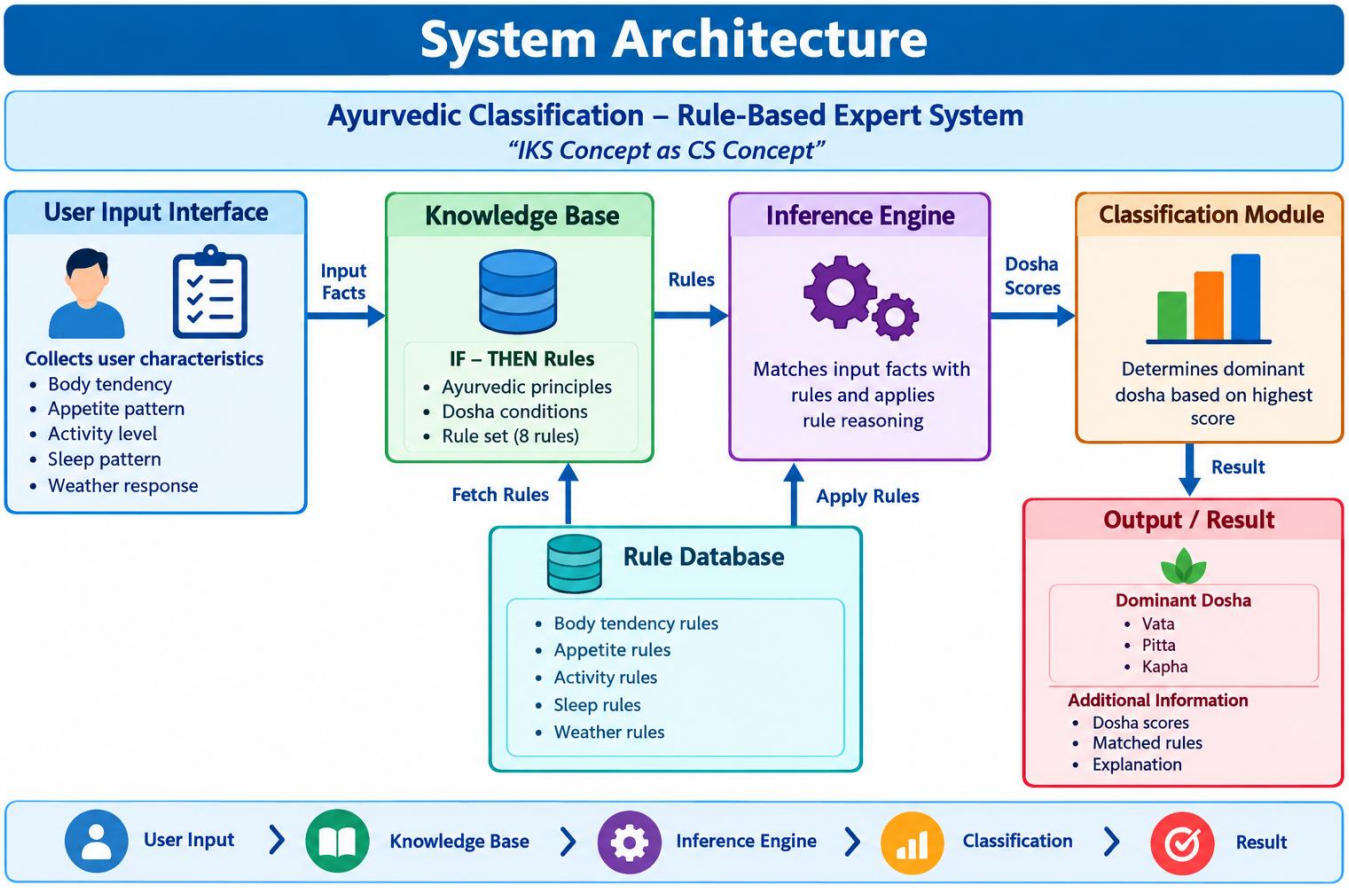
The objective is to demonstrate how traditional knowledge can be represented computationally using **facts, rules, inference, and classification**.

Conceptual Mapping Table

Ayurvedic Classification – Rule-Based Expert System

Sr. No.	Ayurvedic / IKS Concept	Computer Science Concept	System Implementation
1	**Vata, Pitta, Kapha**	Classes / Categories	Three classification categories
2	**Dosha characteristics**	Features / Attributes	Body, appetite, activity, sleep, weather
3	**Ayurvedic principles**	Knowledge Base	Stored as predefined rules
4	**Traditional expert knowledge**	Knowledge Representation	IF–THEN production rules
5	**Ayurvedic classification rules**	Production Rules	Example: IF appetite = strong THEN Pitta
6	**Observation of characteristics**	User Input / Facts	User selects characteristics
7	**Applying Ayurvedic rules**	Inference Engine	Python checks matching rules
8	**Dosha assessment**	Classification	Highest score determines result
9	**Dominant Dosha**	Predicted Class	Vata / Pitta / Kapha
10	**Multiple observations**	Feature Set	Collection of user inputs
11	**Rule matching**	Pattern Matching	Input values are compared with rules
12	**Dosha score**	Classification Score	Each matched rule adds one point
13	**Expert reasoning**	Rule-Based Reasoning	IF–THEN logical inference
14	**Traditional knowledge source**	Database / Rule Repository	Rule dictionary in Python
15	**Final Ayurvedic classification**	System Output	Displays dosha and scores

Basic architecture



Algorithm: Ayurvedic Rule-Based Classification

Objective

The main objective of this project is to classify a user's characteristics into Vata, Pitta, or Kapha using predefined IF–THEN rules.

Input

The system considers the following user characteristics:

- Body tendency
- Appetite pattern
- Activity level
- Sleep pattern
- Weather response

Output

The system provides the following outputs:

- Vata Score
- Pitta Score
- Kapha Score
- Dominant Dosha

Step-by-Step Algorithm

1. Start
2. Initialize three scores:
   - Vata = 0
   - Pitta = 0
   - Kapha = 0
3. Ask the user to enter their body tendency.
4. Apply the corresponding IF–THEN rule:
   - IF body = thin → increase Vata
   - IF body = medium → increase Pitta
   - IF body = solid → increase Kapha
5. Ask the user for their appetite pattern.
6. Apply the corresponding IF–THEN rule and update the appropriate score.
7. Ask for the activity level and apply its rule.
8. Ask for the sleep pattern and apply its rule.
9. Ask for the weather response and apply its rule.
10. Add the scores obtained from all matching rules.
11. Compare the Vata, Pitta, and Kapha scores.
12. If Vata has the highest score, classify the user as Vata.
13. If Pitta has the highest score, classify the user as Pitta.
14. If Kapha has the highest score, classify the user as Kapha.
15. If two or more scores are equal and highest, classify the user as Mixed/Equal.
16. Display the Vata score, Pitta score, Kapha score, matched rules, and final classification.
17. Stop.

Pseudocode

```
START

Vata ← 0
Pitta ← 0
Kapha ← 0

INPUT body

IF body = "thin"
   Vata ← Vata + 1
ELSE IF body = "medium"
   Pitta ← Pitta + 1
ELSE IF body = "solid"
   Kapha ← Kapha + 1

INPUT appetite

IF appetite = "irregular"
   Vata ← Vata + 1
ELSE IF appetite = "strong"
   Pitta ← Pitta + 1
ELSE IF appetite = "steady"
   Kapha ← Kapha + 1

INPUT activity

IF activity = "restless"
   Vata ← Vata + 1
ELSE IF activity = "active"
   Pitta ← Pitta + 1
ELSE IF activity = "slow"
   Kapha ← Kapha + 1

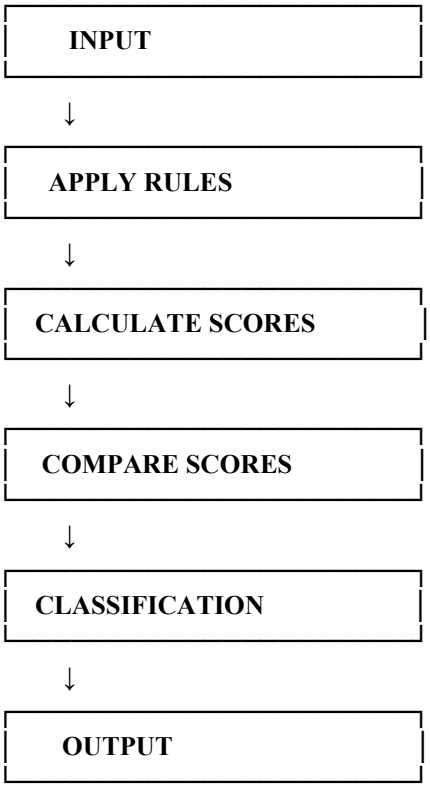
INPUT sleep

IF sleep = "light"
   Vata ← Vata + 1
ELSE IF sleep = "moderate"
   Pitta ← Pitta + 1
ELSE IF sleep = "deep"
   Kapha ← Kapha + 1
```

```
INPUT weather

IF weather = "cold"
   Vata ← Vata + 1
ELSE IF weather = "hot"
   Pitta ← Pitta + 1
ELSE IF weather = "damp"
   Kapha ← Kapha + 1
IF Vata > Pitta AND Vata > Kapha
   RESULT ← "Vata"
ELSE IF Pitta > Vata AND Pitta > Kapha
   RESULT ← "Pitta"
ELSE IF Kapha > Vata AND Kapha > Pitta
   RESULT ← "Kapha"
ELSE
   RESULT ← "Mixed / Equal Classification"
DISPLAY Vata
DISPLAY Pitta
DISPLAY Kapha
DISPLAY RESULT
STOP
```

Flow:



Working Python Code

```
============================================================
# IKS CONCEPT AS CS CONCEPT
# Ayurvedic Classification - Rule-Based Expert System
# Educational Demonstration
============================================================

print("=" * 60)
print(" AYURVEDIC CLASSIFICATION - RULE BASED EXPERT SYSTEM")
print("=" * 60)

# Knowledge Base

rules = {
    "body": {
        "thin": ("Vata", "Thin/light body tendency -> Vata"),
        "medium": ("Pitta", "Medium body tendency -> Pitta"),
        "solid": ("Kapha", "Solid/heavy body tendency -> Kapha")
    },
    "appetite": {
        "irregular": ("Vata", "Irregular appetite -> Vata"),
        "strong": ("Pitta", "Strong appetite -> Pitta"),
        "steady": ("Kapha", "Steady appetite -> Kapha")
    },
    "activity": {
        "restless": ("Vata", "Restless/highly variable activity -> Vata"),
        "active": ("Pitta", "Active and goal-oriented activity -> Pitta"),
        "slow": ("Kapha", "Slow/steady activity -> Kapha")
    },
    "sleep": {
        "light": ("Vata", "Light/irregular sleep -> Vata"),
        "moderate": ("Pitta", "Moderate sleep -> Pitta"),
        "deep": ("Kapha", "Deep/long sleep -> Kapha")
    },
    "weather": {
        "cold": ("Vata", "Discomfort in cold weather -> Vata"),
        "hot": ("Pitta", "Discomfort in hot weather -> Pitta"),
        "damp": ("Kapha", "Discomfort in damp weather -> Kapha")
    }
}
# Function to get valid input
def get_input(question, options):

    print("\n" + question)

    for i, option in enumerate(options, 1):
        print(f"{i}. {option}")

    while True:
```

```
        try:
            choice = int(input("Enter choice: "))

            if 1 <= choice <= len(options):
                return options[choice - 1]

            print("Please enter a valid option.")
        except ValueError:
            print("Please enter a number.")

# Collect user inputs

body = get_input(
    "Select your general body tendency:",
    ["thin", "medium", "solid"]
)
appetite = get_input(
    "Select your appetite pattern:",
    ["irregular", "strong", "steady"]
)
activity = get_input(
    "Select your activity pattern:",
    ["restless", "active", "slow"]
)
sleep = get_input(
    "Select your sleep pattern:",
    ["light", "moderate", "deep"]
)
weather = get_input(
    "Select the weather condition you generally find uncomfortable:",
    ["cold", "hot", "damp"]
)
# Initialize scores
cores = {
    "Vata": 0,
    "Pitta": 0,
    "Kapha": 0
}
atched_rules = []

# Inference Engine

inputs = {
    "body": body,
    "appetite": appetite,
    "activity": activity,
    "sleep": sleep,
    "weather": weather
}
for category, value in inputs.items():
```

```
    dosha, explanation = rules[category][value]

    scores[dosha] += 1

    matched_rules.append(
        f"IF {category} = {value} THEN {dosha}"
    )

# Determine classification

highest_score = max(scores.values())

dominant_doshas = [
    dosha
    for dosha, score in scores.items()
    if score == highest_score
]
if len(dominant_doshas) == 1:
    result = dominant_doshas[0]
else:
    result = "Mixed / Equal Classification"

# Display Results

print("\n" + "=" * 60)
print("EXPERT SYSTEM RESULT")
print("=" * 60)
print("\nInput Facts:")

for key, value in inputs.items():
    print(f"{key.capitalize():12}: {value}")
print("\nDosha Scores:")

for dosha, score in scores.items():
    print(f"{dosha:10}: {score}")

print("\nMatched Rules:")
for rule in matched_rules:
    print(" -", rule)
print("\n" + "-" * 60)
print("CLASSIFICATION:", result)
print("-" * 60)

if result == "Vata":
    print("Vata has the highest rule score.")
elif result == "Pitta":
    print("Pitta has the highest rule score.")
elif result == "Kapha":
    print("Kapha has the highest rule score.")
```

```
else:
    print("Two or more doshas have the same highest score.")
print("\nThis system is an educational demonstration")
print("of rule-based classification and is NOT a medical")
print("diagnosis or treatment system.")
print("=" * 60)
```

Example Input

```
Body tendency : Thin
Appetite      : Irregular
Activity      : Restless
Sleep         : Light
Weather       : Cold
```

Output

```
============================================================
              EXPERT SYSTEM RESULT
============================================================
```

Input Facts:

```
Body        : thin
Appetite    : irregular
Activity    : restless
Sleep       : light
Weather     : cold
```

Dosha Scores:

```
Vata      : 5
Pitta     : 0
Kapha     : 0
```

Matched Rules:

- IF body = thin THEN Vata
- IF appetite = irregular THEN Vata
- IF activity = restless THEN Vata
- IF sleep = light THEN Vata
- IF weather = cold THEN Vata

```
------------------------------------------------------------
CLASSIFICATION: Vata
------------------------------------------------------------

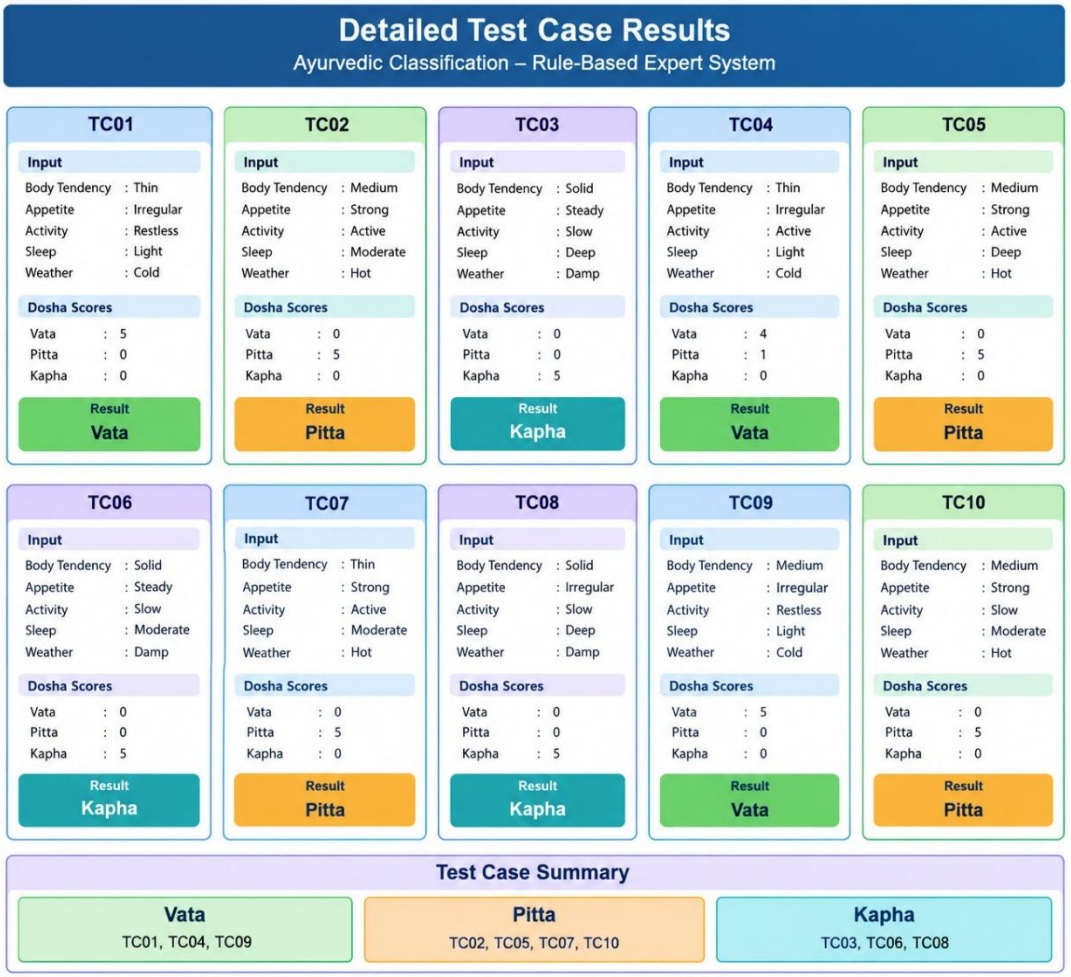
Vata has the highest rule score.

============================================================
```

Test Cases

The following test cases can be included in the **Ayurvedic Classification – Rule-Based Expert System** practical.

Test Case	Body Tendency	Appetite	Activity	Sleep	Weather	Expected Result
TC01	Thin	Irregular	Restless	Light	Cold	**Vata**
TC02	Medium	Strong	Active	Moderate	Hot	**Pitta**
TC03	Solid	Steady	Slow	Deep	Damp	**Kapha**
TC04	Thin	Irregular	Active	Light	Cold	**Vata**
TC05	Medium	Strong	Active	Deep	Hot	**Pitta**
TC06	Solid	Steady	Slow	Moderate	Damp	**Kapha**
TC07	Thin	Strong	Active	Moderate	Hot	**Pitta**
TC08	Solid	Irregular	Slow	Deep	Damp	**Kapha**
TC09	Medium	Irregular	Restless	Light	Cold	**Vata**
TC10	Medium	Strong	Slow	Moderate	Hot	**Pitta**



Complexity & Limitations

The **Ayurvedic Rule-Based Expert System** checks the user's characteristics against a fixed set of predefined rules.

- **Time Complexity (O(n))**

Where **n** = number of input characteristics/rules checked.

For our implementation:

- Body tendency → 1 check
- Appetite → 1 check
- Activity → 1 check
- Sleep → 1 check
- Weather → 1 check

Therefore, the system performs a small, fixed number of operations and gives the result quickly.

- **Space Complexity(O(n))**

Memory is used to store:

- Knowledge-base rules
- User inputs
- Dosha scores
- Matched rules

Limitations

1. **Simplified Knowledge:** The project uses a limited set of Ayurvedic characteristics and does not represent the complete Ayurvedic knowledge system.
2. **Fixed Rules:** The system depends on predefined IF–THEN rules and cannot automatically learn new rules.
3. **Limited Input:** Only five characteristics are considered: body tendency, appetite, activity, sleep, and weather response.
4. **Subjective Answers:** User responses may not always be accurate or consistent.
5. **No Machine Learning:** The system does not improve its classification from previous results or datasets.
6. **Mixed Classification:** When two or more doshas receive the same highest score, the system may return **Mixed/Equal Classification**.
7. **Not a Medical System:** The output is intended only for demonstrating **IKS knowledge representation and rule-based classification**. It should not be used for medical diagnosis or treatment.

Conclusion

The **Ayurvedic Classification – Rule-Based Expert System** successfully demonstrates how an **Indian Knowledge System (IKS)** can be represented using Computer Science concepts.

In this project, Ayurvedic concepts such as **Vata, Pitta, and Kapha** are represented as classification categories, while traditional characteristics are converted into **facts and IF–THEN rules**. The inference engine evaluates the user's inputs, applies the predefined rules, calculates the dosha scores, and produces a classification.

This project demonstrates important concepts of **Artificial Intelligence, knowledge representation, rule-based reasoning, inference engines, and classification**. It also shows how traditional Indian knowledge can be connected with modern computational techniques.

Therefore, the project provides a simple and practical example of converting an IKS concept into a Computer Science rule-based expert system.